Architectural BIM model
Autodesk Revit

USD export

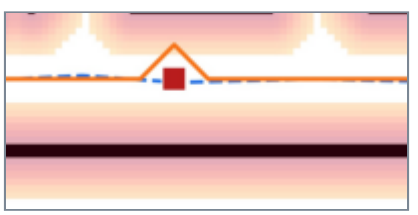
Physics-accurate digital twin
Isaac Sim / USD
colliders, materials,
articulated doors



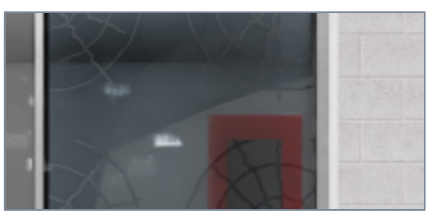
per-twin streams



Routing
occupancy map +
nav graph + TSP route



Local navigation
ROS 2 Nav2 · LiDAR
costmap obstacle avoid.



Perception
Qwen3-VL-2B (LoRA)
synthetic-label anomaly

Autonomous building patrol
Nova Carter · open doors + cracked glass